

AMERICAN INTERNATIONAL UNIVERSITY BANGLADESH
(AIUB)

FACULTY OF SCIENCE & TECHNOLOGY



Course Title
INTRODUCTION TO DATABASE (2108)

Semester: Spring 2023-2024

Section: [O]

TITLE
Company Structure Management System

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1.Introduction

A proficient company management system is fundamental for optimizing operations, coordinating resources, and amplifying productivity. At the heart of such a system lies a meticulously crafted database, pivotal for storing, retrieving, and managing diverse company data facets. These include employee particulars, project specifics, financial records, and beyond. Through the implementation of a robust database management system, enterprises can elevate efficiency, streamline decision-making processes, and uphold data integrity. This introduction aims to elucidate the pivotal components and advantages inherent in a comprehensive company management system driven by a proficient database infrastructure.

In addition to operational benefits, a well-structured company management system powered by a database fosters collaboration and communication among team members. With centralized access to relevant information, employees can collaborate more effectively, share knowledge, and coordinate tasks seamlessly across departments and teams.

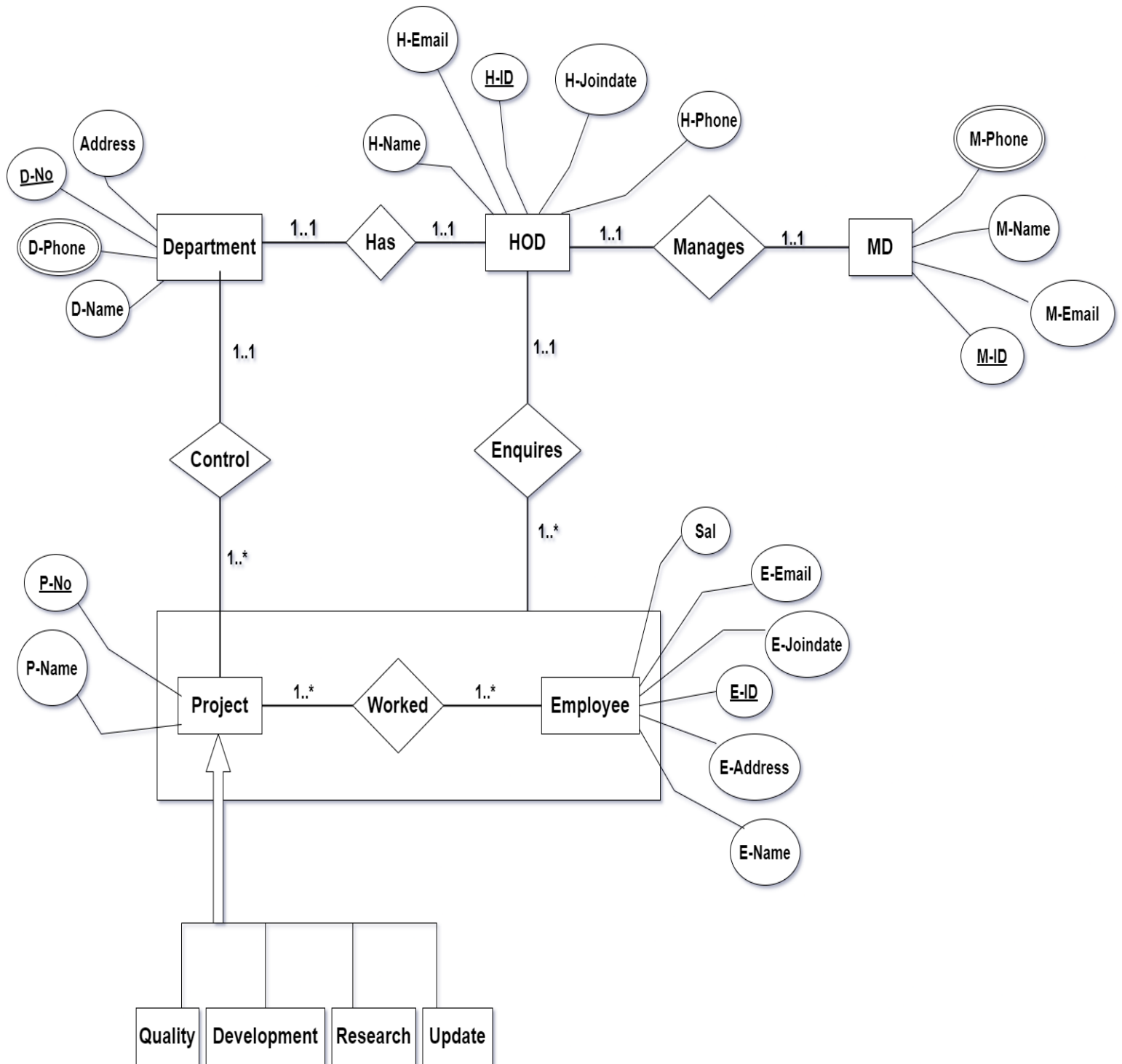
In summary, a comprehensive company management system supported by a robust database infrastructure is indispensable for modern businesses seeking to optimize their operations, make informed decisions, and stay competitive in today's dynamic marketplace.

2.Case Study / Scenario

StudentID1: 22-49496-3 Name: Shuvra Kanti Das	StudentID3: 22-49509-3 Name: Chandrima Dam
StudentID2: 22-49847-3 Name: Sagor Chandra	StudentID4: 22-49828-3 Name: Mst. Fareha Rahman
CO2: Understand the fundamental concepts underlying database systems and gain hands-on experience with ER diagram Case study	
PO-c2: Develop process for complex computer science and engineering problems considering cultural and societal factors.	Marks

In a company structure management system, the company is organized into departments. Each department is identified by a department ID number. The system also stores department name, department email address, and multiple phone numbers for each department. Each department has a Head of Department (HOD). The information of the HOD includes name, id, phone number, join date, address, and salary. The HOD is managed by a Managing Director (MD), who has information such as M-Name, M-email, M-mobile-no, and M-ID. Each department controls several projects aimed at improving the company. Each project has a p-name, and a p-No. Projects are specialized in development, research, quality, and updates. Every project is worked on by many employees. To identify an employee, the system stores E-ID along with E-email, E-address, E-salary, and E-Join-date. The Head of the Department inquiries about projects and employees.

3.ER Diagram



4.Normalization

HAS RELATION:

UNF: D-NAME, D-NO, ADDRESS, D-PHONE, H-ID, H-NAME, H-EMAIL, H-JOINDATE, H-PHONE.

1NF: D-NO(PK), D-NAME, ADDRESS, D-PHONE, H-ID(PK), H-NAME, H-EMAIL, H-JOINDATE, H-PHONE.

2NF:

- i. D-NO(PK), D-NAME, ADDRESS, D-PHONE.
- ii. H-ID(PK), H-NAME, H-EMAIL, H-JOINDATE, H-PHONE.
- iii. D-NO(PK), H-ID(FK).

3NF: As same as 2NF.

MANAGE RELATION:

UNF: H-ID, H-EMAIL, H-JOINDATE, H-NAME, H-PHONE, M-ID, M-NAME, M-EMAIL, M-PHONE.

1NF: H-ID(PK), H-EMAIL, H-JOINDATE, H-NAME, H-PHONE, M-ID(PK), M-NAME, M-EMAIL, M-PHONE.

2NF:

- i. H-ID(PK), H-NAME, H-EMAIL, H-JOINDATE, H-PHONE.
 - ii. M-ID(PK), M-NAME, M-EMAIL, M-PHONE.
 - iii. M-ID(PK), H-ID(FK).
- 3NF: As same as 2NF.

CONTROL RELATION:

UNF: D-NAME, D-NO, ADDRESS, D-PHONE, P-NO, P-NAME.

1NF: D-NO(PK), D-NAME, ADDRESS, D-PHONE, P-NO(PK), P-NAME.

2NF:

- i. D-NO(PK), D-NAME, ADDRESS, D-PHONE.
- ii. P-NO(PK), P-NAME, D-NO(FK).

3NF: As same as 2NF.

WORK RELATION:

UNF: P-NO, P-NAME, E-ID, E-EMAIL, E-NAME, E-JOINDTAE, E-ADDRESS, SAL.

1NF: P-NO(PK), P-NAME, E-ID(PK), E-EMAIL, E-NAME, E-JOINDTAE, E-ADDRESS, SAL.

2NF:

- i. P-NO(PK), P-NAME.
- ii. E-ID(PK), E-EMAIL, E-NAME, E-JOINDTAE, E-ADDRESS, SAL.
- iii. P-NO(PK), E-ID(FK).

3NF: As same as 2NF.

ENQUIRE RELATION:

UNF: H-ID, H-EMAIL, H-JOINDATE, H-NAME, H-PHONE, P-NO, P-NAME, E-ID, E-EMAIL, E-NAME, E-JOINDTAE, E-ADDRESS, SAL.

1NF: H-ID(PK), H-EMAIL, H-JOINDATE, H-NAME, H-PHONE, P-NO(PK), P-NAME, E-ID(PK), E-EMAIL, E-NAME, E-JOINDTAE, E-ADDRESS, SAL.

2NF:

- i. H-ID(PK), H-EMAIL, H-JOINDATE, H-NAME, H-PHONE
- ii. P-NO(FK), P-NAME, E-ID(PK), E-EMAIL, E-NAME, E-JOINDTAE, E-ADDRESS, SAL, H-ID(FK)

3NF: As same as 2NF

5.Finalization

- I. D-NO(PK), D-NAME, ADDRESS, D-PHONE.
- II. H-ID(PK), H-NAME, H-EMAIL, H-JOINDATE, H-PHONE.
- III. D-NO(PK), H-ID(FK).
- IV. H-ID(PK), H-NAME, H-EMAIL, H-JOINDATE, H-PHONE.
- V. M-ID(PK), M-NAME, M-EMAIL, M-PHONE.
- VI. M-ID(PK), H-ID(FK).
- VII. D-NO(PK), D-NAME, ADDRESS, D-PHONE.
- VIII. P-NO(PK), P-NAME, D-NO(FK).
- IX. P-NO(PK), P-NAME

- X. E-ID(PK), E-EMAIL, E-NAME, E-JOINDTAE, E-ADDRESS, SAL.
- XI. P-NO(PK), E-ID(FK).
- XII. H-ID(PK), H-EMAIL, H-JOINDATE, H-NAME, H-PHONE
- XIII. P-NO(FK), P-NAME, E-ID(PK), E-EMAIL, E-NAME, E-JOINDTAE, E-ADDRESS, SAL, H-ID(FK).

5.1 **FINAL TABLES**

- I. D-NO(PK), D-NAME, ADDRESS, D-PHONE. **(DEPARTMENT)**
- II. D-NO(PK), H-ID(FK). **(HAS)**
- III. M-ID(PK), M-NAME, M-EMAIL, M-PHONE. **(MANAGER)**
- IV. M-ID(PK), H-ID(FK). **(MANAGE)**
- V. P-NO(PK), P-NAME, D-NO(FK). **(CONTROL)**
- VI. P-NO(PK), P-NAME. **(PROJECT)**
- VII. E-ID(PK), E-EMAIL, E-NAME, E-JOINDTAE, E-ADDRESS, SAL. **(EMPLOYEE)**
- VIII. P-NO(PK), E-ID(FK). **(WORK)**
- IX. H-ID(PK), H-EMAIL, H-JOINDATE, H-NAME, H-PHONE. **(HOD)**
- X. P-NO(FK), P-NAME, E-ID(PK), E-EMAIL, E-NAME, E-JOINDTAE, E-ADDRESS, SAL, H-ID(FK). **(ENQUIRE)**

5.2 Final Table Name

- i. Department
- ii. Has
- iii. Manager
- iv. Manage
- v. Control
- vi. Project
- vii. Employee
- viii. Work
- ix. Hod
- x. Enquire

6. Table Creation (DDL Operations)

StudentID1: 22-49496-3 Name: Shuvra Kanti Das	StudentID3: 22-49509-3 Name: Chandrima Dam
StudentID2: 22-49847-3 Name: Sagor Chandra	StudentID4: 22-49828-3 Name: Mst. Fareha Rahman
CO4: Creating DML, DDL using Oracle and connection with ODBC/JDBC for existing JAVA application	
PO-e-2: Use modern engineering and IT tools for prediction and modeling of complex computer science and engineering problem	Marks

6.1 Creation Of Department Table:

User: COMPANYMNGT

Home > SQL > SQL Commands

☒ Autocommit Display 100 ▼

```
create table department (DNo number (2) primary key,  
DName varchar (10), DAdress varchar (20), DPhn number (15))
```

FIGURE – 6.1

6.2 Describe Of Department:

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>DEPARTME</u> <u>NT</u>	<u>DNO</u>	Number	-	2	0	1	-	-	-
	<u>DNAM</u> <u>E</u>	Varchar2	10	-	-	-	✓	-	-
	<u>DADR</u> <u>ESS</u>	Varchar2	20	-	-	-	✓	-	-
	<u>DPHN</u>	Number	-	15	0	-	✓	-	-
1 - 4									

FIGURE – 6.2

6.3 Creation Of Has Table:

User: COMPANYMNGT

Home > SQL > **SQL Commands**

☒ Autocommit Display 100 v

```
create table has (DNo number (2) primary key, HID number (2),  
constraint hd foreign key (HID) references HOD (HID))
```

FIGURE – 6.3

6.4 Describe Of Has Table:

Table	C o l u m n	Data Type	Length	Precision	Scale	Primary Key	Nullabl e	Default	Comme nt
<u>HAS</u>	<u>D</u> <u>N</u> <u>O</u>	Number	-	2	0	1	-	-	-
	<u>H</u> <u>I</u> <u>D</u>	Number	-	2	0	-	✓	-	-
1 - 2									

FIGURE – 6.4

6.5 Creation Of Manager Table:

User: COMPANYMNGT

Home > SQL > **SQL Commands**

☒ Autocommit Display 100 ▾

```
create table manager (MID number (2) primary key,  
Mname varchar (10), Mmail varchar (12), MPhn number (15))
```

FIGURE – 6.5

6.6 Describe Of Manager Table:

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>MANAGER</u>	<u>MID</u>	Number	-	2	0	1	-	-	-
	<u>MNAME</u>	Varchar2	10	-	-	-	✓	-	-
	<u>MMAIL</u>	Varchar2	12	-	-	-	✓	-	-
	<u>MPHN</u>	Number	-	15	0	-	✓	-	-

1 - 4

FIGURE – 6.6

6.7 Creation Of Manage Table:

User: COMPANYMNGT

Home > SQL > **SQL Commands**

☒ Autocommit Display 100 ▼

```
create table manage (MID number (2) primary key,  
HID number (2), constraint mh foreign key (HID) references HOD (HID))
```

FIGURE – 6.7

6.8Describe Of Manage Table:

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>MANAGE</u>	<u>MID</u>	Number	-	2	0	1	-	-	-
	<u>HID</u>	Number	-	2	0	-	✓	-	-

1 - 2

FIGURE – 6.8

6.9 Creation Of Control Table:

User: COMPANYMGT

Home > SQL > SQL Commands

☒ Autocommit Display 100 ▼

```
create table control (PNo number (2) primary key, PName varchar (10),  
DNo number (2), constraint dp foreign key (DNo) references department (DNo))
```

FIGURE – 6.9

6.10 Describe Of Control Table:

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>CONTROL</u>	<u>PNO</u>	Number	-	2	0	1	-	-	-
	<u>PNAME</u>	Varchar2	10	-	-	-	✓	-	-
	<u>DNO</u>	Number	-	2	0	-	✓	-	-

1 - 3

FIGURE – 6.10

6.11 Creation Of Project Table:

User: COMPANYMNGT

Home > SQL > **SQL Commands**

☒ Autocommit Display 100 ▾

```
create table project (PNo number (2) primary key,  
PName varchar (10))
```

FIGURE – 6.11

6.12 Describe Of Project Table:

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>PROJECT</u>	<u>PNO</u>	Number	-	2	0	1	-	-	-
	<u>PNAME</u>	Varchar2	10	-	-	-	✓	-	-

1 - 2

FIGURE – 6.12

6.13 Creation Of Employee Table:

User: COMPANYMNGT

Home > SQL > SQL Commands

☒ Autocommit Display 100 ▼

```
create table employee (EID number (2)primary key, Ejoindate date,
EName varchar (10), Email varchar (12), Esal number (8), EAddress varchar (20))
```

FIGURE – 6.13

6.14 Describe Of Employee Table:

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>EMPLOYEE</u>	<u>EID</u>	Number	-	2	0	1	-	-	-
	<u>EJOINDATE</u>	Date	7	-	-	-	✓	-	-
	<u>ENAME</u>	Varchar2	10	-	-	-	✓	-	-
	<u>EMAIL</u>	Varchar2	12	-	-	-	✓	-	-
	<u>ESAL</u>	Number	-	8	0	-	✓	-	-
	<u>EADDRESS</u>	Varchar2	20	-	-	-	✓	-	-

1 - 6

FIGURE – 6.14

6.15 Creation Of Work Table:

User: COMPANYMNGT

Home > SQL > SQL Commands

☒ Autocommit Display 100 ▼

```
create table work (PNo number (2) primary key, EID number (2),  
constraint ep foreign key (EID) references employee (EID))
```

FIGURE – 6.15

6.16 Describe Of Work Table:

Table	Col umn	Data Type	Lengt h	Precisio n	Scal e	Primary Key	Nullab le	Defau lt	Comme nt
<u>WORK</u>	<u>P N O</u>	Number	-	2	0	1	-	-	-
	<u>EID</u>	Number	-	2	0	-	✓	-	-
1 - 2									

FIGURE – 6.16

6.17 Creation Of HOD Table:

User: COMPANYMNGT

Home > SQL > SQL Commands

☒ Autocommit Display 100 ▼

```
create table HOD (HID number (2) primary key,  
HName varchar (10), Hmail varchar (12), HPhn number (15), Hjoindate date)
```

FIGURE – 6.17

6.18 Describe Of HOD Table:

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>HOD</u>	<u>HID</u>	Number	-	2	0	1	-	-	-
	<u>HNAME</u>	Varchar2	10	-	-	-	✓	-	-
	<u>HMAIL</u>	Varchar2	12	-	-	-	✓	-	-
	<u>HPHN</u>	Number	-	15	0	-	✓	-	-
	<u>HJOINDATE</u>	Date	7	-	-	-	✓	-	-

1 - 5

FIGURE – 6.18

6.19 Creation Of Enquire Table:

User: COMPANYMNGT

Home > SQL > SQL Commands

☒ Autocommit Display 100

```
create table enquire (HID number (2), EID number (2) primary key,
EName varchar (10),Email varchar (12), Esal number (8),
EAdress varchar (20), PNo number (2), constraint ghk foreign key (PNo)
references project (PNo), constraint phk foreign key (HID) references HOD (HID))
```

FIGURE – 6.19

6.20 Describe Of Enquire Table:

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>ENQUIRE</u>	<u>HID</u>	Number	-	2	0	-	✓	-	-
	<u>EID</u>	Number	-	2	0	1	-	-	-
	<u>ENAME</u>	Varchar2	10	-	-	-	✓	-	-
	<u>EMAIL</u>	Varchar2	12	-	-	-	✓	-	-
	<u>ESAL</u>	Number	-	8	0	-	✓	-	-
	<u>EADDRESS</u>	Varchar2	20	-	-	-	✓	-	-
	<u>PNO</u>	Number	-	2	0	-	✓	-	-
1 - 7									

FIGURE – 6.20

7.Inserted Values in the table

7.1 Values Of Department Table :

Results	Explain	Describe	Saved SQL	History
DNO	DNAME	DADDRESS	DPHN	
10	Marketing	Dhaka	783250936	
20	IT	Comilla	783250946	
30	HR	Narayanganj	783250938	
40	Management	Kulna	783259936	
50	Finance	Sylhet	757353226	

5 rows returned in 0.00 seconds [CSV Export](#)

FIGURE – 7.1

7.4 Values Of Has Table :

Results	Explain	Describe	Saved SQL	History
DNO	HID			
10	15			
20	25			
30	35			
40	75			

4 rows returned in 0.00 seconds [CSV Export](#)

FIGURE – 7.2

7.3 Values Of Manager Table :

Results	Explain	Describe	Saved SQL	History
MID	MNAME	MMAIL	MPHN	
11	TRVIS	tr@gmail.com	1677638822	
22	PERRY	pr@gmail.com	1677638822	
33	JOHN	jh@gmail.com	1677638822	
44	MACC	ma@gmail.com	1677638822	
4 rows returned in 0.00 seconds				CSV Export

FIGURE – 7.3

7.8 Values Of Manage Table :

Results	Explain	Describe	Saved SQL	History
MID	HID			
11	15			
22	25			
33	75			
44	35			
4 rows returned in 0.00 seconds				CSV Export

FIGURE – 7.4

7.5 Values Of Control Table :

Results	Explain	Describe	Saved SQL	History
PNO	PNAME	DNO		
10	Local Link	10		
20	Unity Hub	20		
30	Safe Zone	30		
40	EcoPower	40		
50	Urban Link	50		

5 rows returned in 0.00 seconds [CSV Export](#)

FIGURE – 7.5

7.6 Values Of Project Table :

Results	Explain	Describe	Saved SQL	History
PNO	PNAME			
50	Urban Link			
10	Local Link			
20	Unity Hub			
30	Safe Zone			
40	EcoPower			

5 rows returned in 0.00 seconds [CSV Export](#)

FIGURE – 7.6

7.7 Values Of Employee Table :

Results Explain Describe Saved SQL History

EID	EJOINDATE	ENAME	EMAIL	ESAL	EADDRESS
30	01-MAR-08	ALEX CARRY	x@gmail.com	12000	BARISAL
40	23-SEP-18	MARK WOOD	a@gmail.com	30000	MOULVIBAZAR
50	10-AUG-09	TIM DAVID	t@gmail.com	4000	CHITTAGONG
60	17-NOV-15	MARRY COM	m@gmail.com	75000	MYMENSINGH
70	30-APR-23	PAT MILLS	p@gmail.com	6000	GAZIPUR

5 rows returned in 0.00 seconds

[CSV Export](#)

FIGURE – 7.7

7.8 Values Of Work Table :

Results Explain Describe Saved SQL History

PNO	EID
10	30
20	40
30	50
40	60
50	70

5 rows returned in 0.00 seconds

[CSV Export](#)

FIGURE – 7.8

7.9 Values Of HOD Table :

Results	Explain	Describe	Saved SQL	History
HID	HNAME	HMAIL	HPHN	HJOINDATE
75	CHANDRIMA	cd@yahoo.com	98003776383	12-FEB-02
25	FAREHA	fr@gmail.com	90763462163	30-MAR-14
35	SHUVRO	sk@gmail.com	98078350098	21-MAY-10
15	SAGOR	cd@gmail.com	95735342773	01-FEB-08

4 rows returned in 0.02 seconds [CSV Export](#)

FIGURE – 7.9

7.10 Values Of Enquire Table :

ResultsExplainDescribeSaved SQLHistory

HID	EID	ENAME	EMAIL	ESAL	EADDRESS	PNO
15	30	ALEX CARRY	x@gmail.com	12000	BARISAL	10
25	40	MARK WOOD	a@gmail.com	30000	MOULVIBAZAR	20
35	50	TIM DAVID	t@gmail.com	4000	CHITTAGONG	30
75	60	MARRY COM	m@gmail.com	75000	MYMENSINGH	40

4 rows returned in 0.00 seconds

CSV Export

FIGURE – 7.10

8. Query Test

8.1 Simple Query

➔ Show department name and department number from the department table.

User: COMPANYMNGT

Home > SQL > **SQL Commands**

☒ Autocommit Display 100 ▼

```
select dname, dno from department
```

FIGURE – 8.1

Results	Explain	Describe	Saved SQL	History
DNAME	DNO			
Marketing	10			
IT	20			
HR	30			
Management	40			
Finance	50			

5 rows returned in 0.02 seconds [CSV Export](#)

FIGURE – 8.2

8.2 Single Row Function

➔ Show the ID, name, and email address for managers named 'John'.

User: COMPANYMNGT

Home > SQL > **SQL Commands**

☒ Autocommit Display 10 ▼

```
SELECT Mid, Mname, Mmail
FROM   manager
WHERE  LOWER(Mname) = 'john'
```

FIGURE – 8.3

Results Explain Describe Saved SQL History

MID	MNAME	MMAIL
33	JOHN	jh@gmail.com

1 rows returned in 0.00 seconds

CSV Export

FIGURE – 8.4

8.3 Multiple Row Function

- ➔ Show the names, the length of the names and the position of the first occurrence of the letter 'A' in the names from the enquire table.

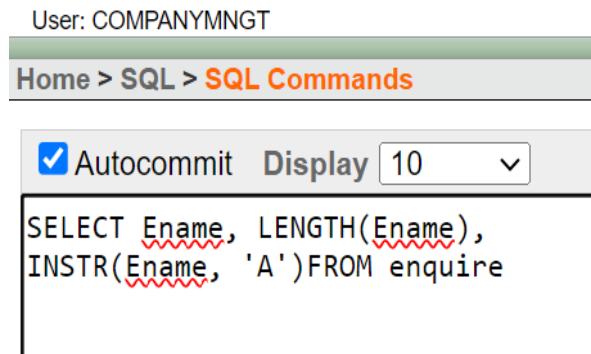


FIGURE – 8.5

Results	Explain	Describe	Saved SQL	History
ENAME	LENGTH(ENAME)	INSTR(ENAME,'A')		
ALEX CARRY	10	1		
MARK WOOD	9	2		
TIM DAVID	9	6		
MARRY COM	9	2		

4 rows returned in 0.02 seconds [CSV Export](#)

FIGURE – 8.6

8.4 Group Function

➔ Show maximum salary and minimum salary from enquire table.

User: COMPANYMNGT

Home > SQL > **SQL Commands**

☒ Autocommit Display ▼

```
select MIN(Esal), MAX(Esal) from enquire
```

FIGURE – 8.7

Results	Explain	Describe	Saved SQL	History				
<table><tr><th>MIN(ESAL)</th><th>MAX(ESAL)</th></tr><tr><td>4000</td><td>75000</td></tr></table>					MIN(ESAL)	MAX(ESAL)	4000	75000
MIN(ESAL)	MAX(ESAL)							
4000	75000							
1 rows returned in 0.00 seconds			CSV Export					

FIGURE – 8.8

8.5 Single Row Subquery

➔ Show the ename and joindate from employee table whose esal is less than eid = 70

User: COMPANYMGT

Home > SQL > SQL Commands

☒ Autocommit Display 10 ▼

```
SELECT ename, ejointdate
FROM employee
WHERE Esal < (SELECT Esal FROM employee WHERE EID = 70);
```

FIGURE – 8.9

Results Explain Describe Saved SQL History

ENAME	EJOINDATE
TIM DAVID	10-AUG-09

1 rows returned in 0.00 seconds

CSV Export

FIGURE – 8.10

8.6 Multiple Row Subquery

- ➔ Show the eid and joindate from employee table where esal less than all of the 'ALEX CARRY'.

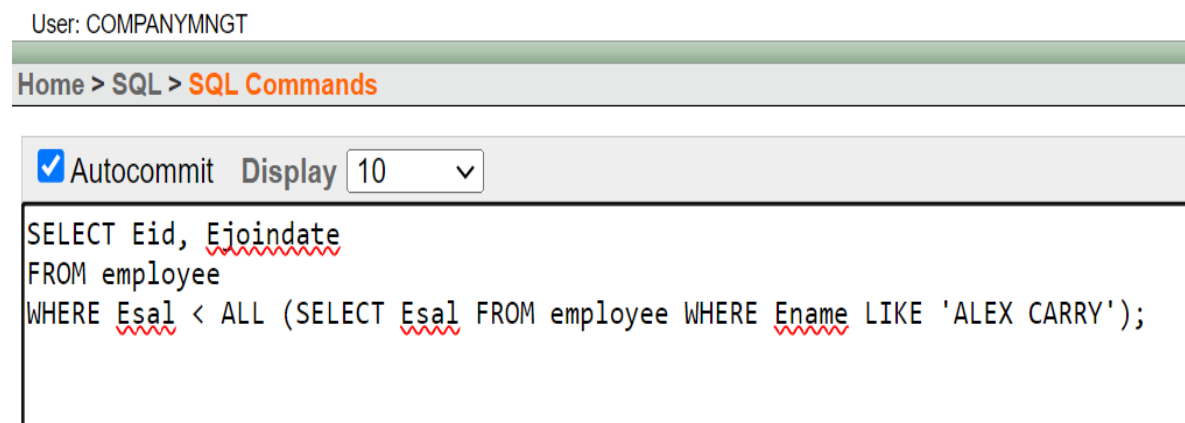


FIGURE – 8.11

Results	Explain	Describe	Saved SQL	History
EID	EJOINDATE			
50	10-AUG-09			
70	30-APR-23			

2 rows returned in 0.02 seconds [CSV Export](#)

FIGURE – 8.12

8.7 Self Join

- ➔ Show a self-join in the Department table where it will show name and address within itself.

User: COMPANYMNGT

Home > SQL > SQL Commands

☒ Autocommit Display 50 ▼

```
SELECT a.dname||' department is located in '||b.daddress
FROM   department a, department b
WHERE  a.dno = b.dno;
```

FIGURE – 8.13

Results	Explain	Describe	Saved SQL	History
A.DNAME 'DEPARTMENTISLOCATEDIN' B.DADDRESS				
Marketing department is located in Dhaka				
IT department is located in Comilla				
HR department is located in Narayanganj				
Management department is located in Kulna				
Finance department is located in Sylhet				
5 rows returned in 0.01 seconds CSV Export				

FIGURE – 8.14

8.8 Outer Join

- ➔ Show pname, pno, eid from the project and work table according to the order (default) of pno.

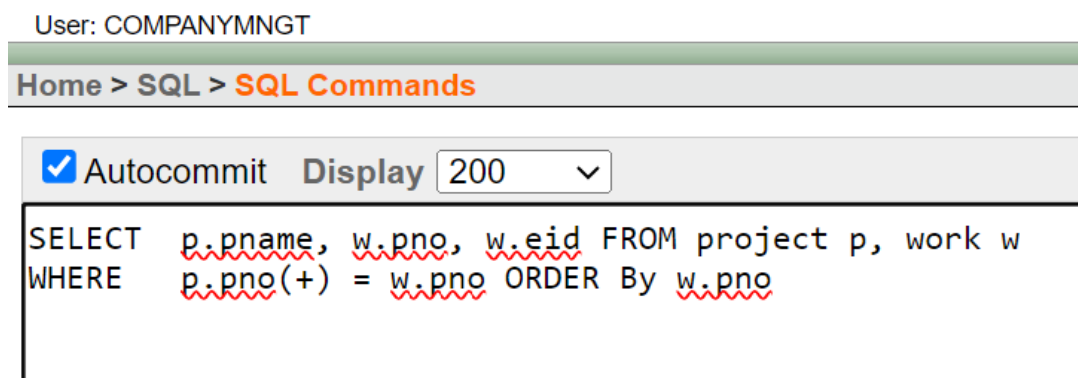


FIGURE – 8.15

Results	Explain	Describe	Saved SQL	History
PNAME	PNO	EID		
Local Link	10	30		
Unity Hub	20	40		
Safe Zone	30	50		
EcoPower	40	60		
Urban Link	50	70		

5 rows returned in 0.01 seconds [CSV Export](#)

FIGURE – 8.16

8.9 Simple View

- ➔ Create a view named as **hodvu30** and show the id, name, join date of the HOD where hid is 15.

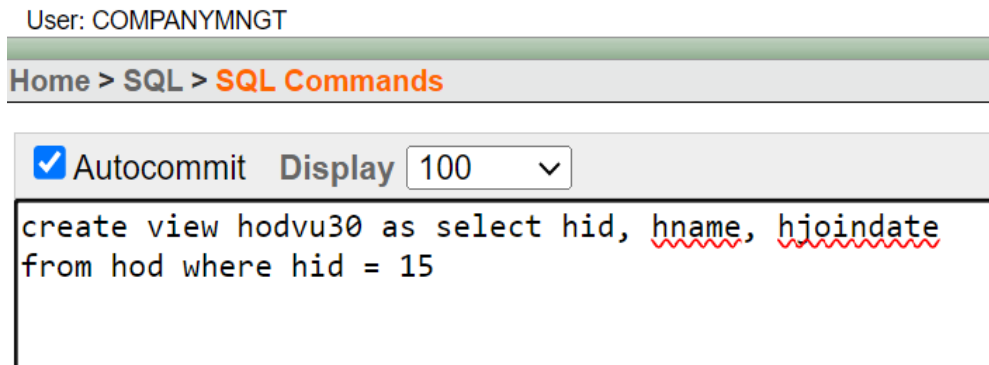


FIGURE – 8.17

Results Explain Describe Saved SQL History									
Object Type VIEW Object HODVU30									
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
HODVU30	HID	Number	-	2	0	-	-	-	-
	HNAME	Varchar2	10	-	-	-	✓	-	-
	HJOINDATE	Date	7	-	-	-	✓	-	-
1 - 3									

FIGURE – 8.18: Description of the simple view

Results	Explain	Describe	Saved SQL	History
HID	HNAME	HJOINDATE		
15	SAGOR	01-FEB-08		

1 rows returned in 0.00 seconds [CSV Export](#)

FIGURE – 8.19: Simple view as a whole table

8.10 Complex View

- ➔ Create a view named compview10 (SUMSAL) as select sum (esal) from enquire where ename like '%A%'.

User: COMPANYMNGT

Home > SQL > **SQL Commands**

☒ Autocommit Display 100 ▾

```
create view compview10 (sumsal) as select sum(esal)
FROM enquire WHERE ename like '%A%'
```

FIGURE – 8.20

Results Explain Describe Saved SQL History

Object Type VIEW Object COMPVIEW10

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
COMPVIEW10	SUMSAL	Number	-	-	-	-	✓	-	-
									1 - 1

FIGURE – 8.21: Description of the complex view

Results	Explain	Describe	Saved SQL	History
SUMSAL				
121000				
1 rows returned in 0.00 seconds				
CSV Export				

FIGURE – 8.22: Complex view as a whole table

9. DB Connection

DB Connection -1 :

SAGOR CHANDRA (Table Name: **Manager**)\

- 1.First, install the MySQL Java Connector by downloading the Maven package containing the JAR file version 8.4.0.
- 2.Download and install XAMPP, which includes Apache, MariaDB, Perl, PHP, and MySQL servers. Start the Apache and MySQL servers, then open the MySQL server admin panel.
- 3.In the admin panel, create a database called "companymngt" and a table named "manager" with four columns. Insert values into the table, adding five rows in total.

Here is MySQL table

MID	MName	Mmail	Mphn
11	TRAVIS	tr@gmail.com	1677638822
22	PERRY	pr@gmail.com	1677638822
33	JOHN	jh@gmail.com	1677638822
44	MACC	ma@gmail.com	1677638822

FIGURE – 9.1 Output of the table from the MySQL server.

4.Download and install the Apache NetBeans IDE21. Then, add the JAR file to the project's library section in the IDE.

5.Write the DB connection code inside the "manager" class of the project. The code involves:

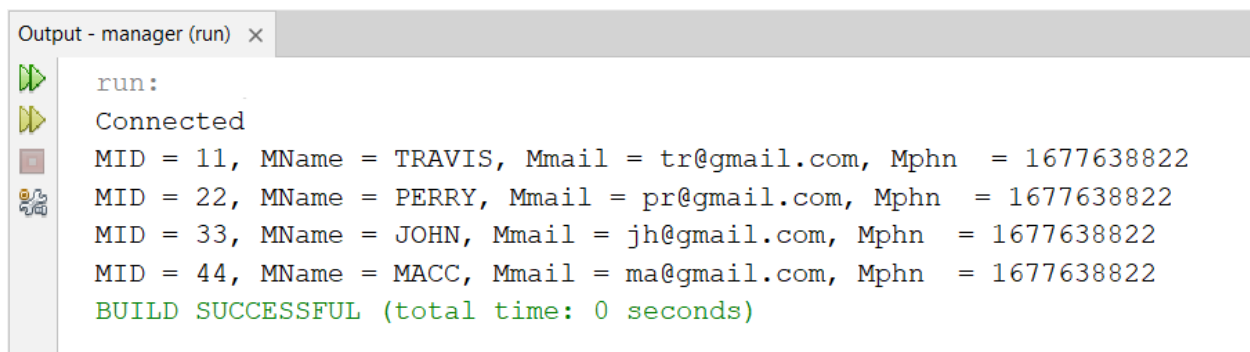
- ❖ Registering the MySQL JDBC driver dynamically using `Class.forName("com.mysql.cj.jdbc.Driver")`. JDBC drivers are essential for connecting Java applications with databases.
- ❖ Establishing a database connection using the `DriverManager` and obtaining a connection object.
- ❖ Creating a statement object using the connection's `createStatement` method.
- ❖ Executing SQL queries and obtaining result sets using the statement object.
- ❖ Closing the database connection using the `connection.close()` method for proper resource management.
- ❖ The **Manager** class is run.

The Main Code –

```
1 import java.sql.*;
2 public class Manager {
3     public static void main(String[] args) {
4         try {
5             Class.forName("com.mysql.cj.jdbc.Driver");
6             Connection connection = DriverManager.getConnection("jdbc:mysql://localhost:3306/companymngt", "root", "");
7             System.out.println("Connected");
8             Statement statement = connection.createStatement();
9             ResultSet resultSet = statement.executeQuery("SELECT * FROM `manager`");
10            while (resultSet.next()) {
11                System.out.print("MID = " + resultSet.getInt(1) + ", ");
12                System.out.print("MName = " + resultSet.getString(2) + ", ");
13                System.out.print("Mmail = " + resultSet.getString(3) + ", ");
14                System.out.println("Mphn = " + resultSet.getInt(4));
15            }
16            connection.close();
17        } catch (Exception ex) {
18            System.out.println(ex);
19        }
20    }
21 }
22 }
```

FIGURE – 9.2 Java code to connect MySQL database and to show the output.

The Output-



```
run:
Connected
MID = 11, MName = TRAVIS, Mmail = tr@gmail.com, Mphn = 1677638822
MID = 22, MName = PERRY, Mmail = pr@gmail.com, Mphn = 1677638822
MID = 33, MName = JOHN, Mmail = jh@gmail.com, Mphn = 1677638822
MID = 44, MName = MACC, Mmail = ma@gmail.com, Mphn = 1677638822
BUILD SUCCESSFUL (total time: 0 seconds)
```

FIGURE – 9.3 Output of the query from the code.

DB Connection -2 :

Fareha Rahman (Table Name : **Department**)

1. First go to MySQL java connector file .
2. Then search and install xampp . After installed we have to go xampp control and start Apache and MySQL.
3. Go to localhost and create a new table with table name and insert the values in the table.

The table in MySQL:

DNO	DNAME	DADDRESS	DPHONE
10	Marketing	Dhaka	783250936
20	IT	Comilla	783250946
30	HR	Narayanganj	783250938
40	Management	Kulna	783259936
50	Finance	Sylhet	757353226

FIGURE – 9.4 Output of the table from the MySQL server.

4.Chosen a Integrated Development Environment (IDE) of my choice. I used Netbense code for Java development.

5. In my Java code, loaded MySQL. Java Connector driver by imporing the JAR file into my project and using the Class.forName() method to register it.

6.Used the DriverManager.getConnection() method to establish a connection to my MySQL database by providing the appropriate URL, username, and password. My usernanie root and there was no password set.

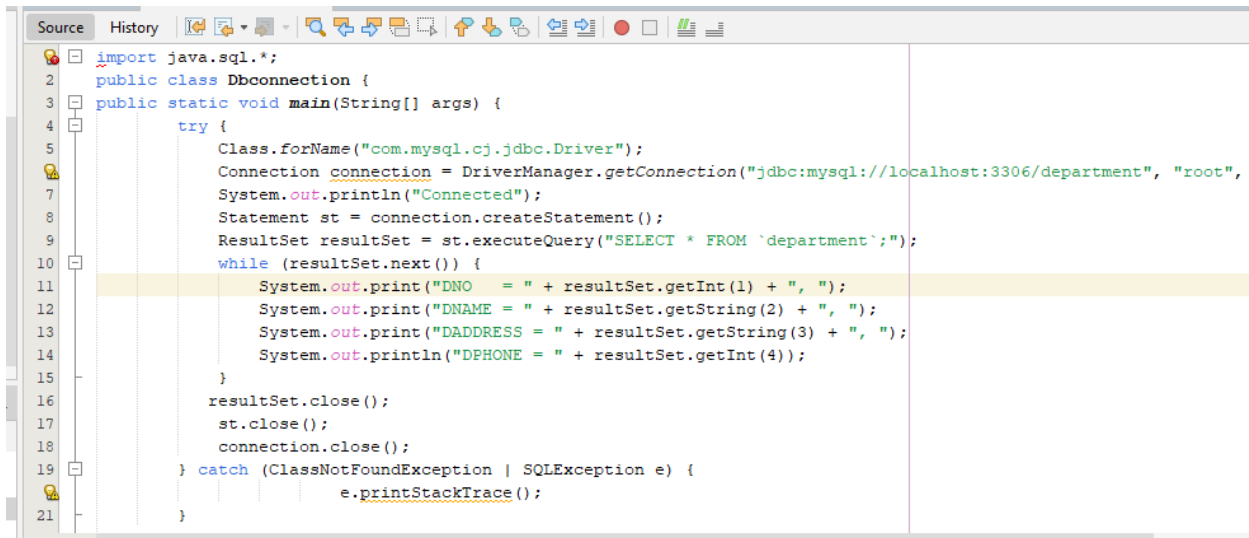
7.After establishing the connection, created a Statement or CrenetedStatement object to execute SQL queries .

8.Used the crusted statement 10 exusate SOL queries like SELECT, INSERT,UPDATE, or DELETE. Captured the necessary results.

9.Used the ResultSet object to retrieve and process the data

10.Performed necessary operations on the dats retrieved from the database.

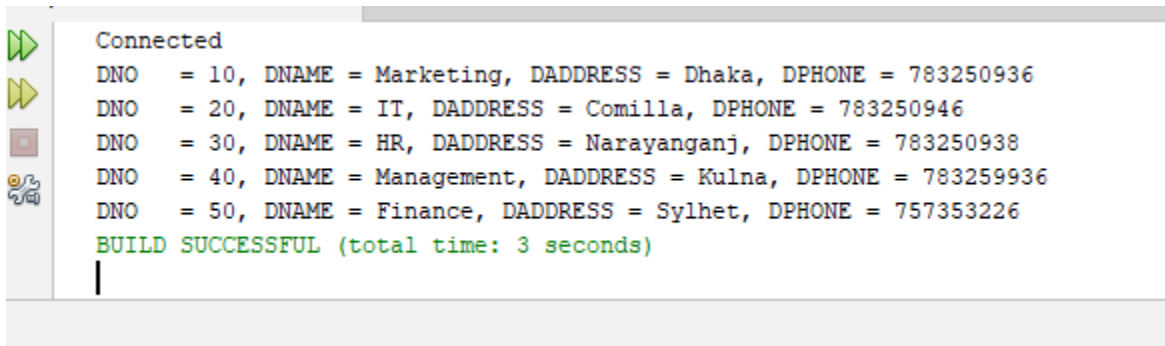
The Main Code –



```
Source History
import java.sql.*;
public class Dbconnection {
public static void main(String[] args) {
    try {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection connection = DriverManager.getConnection("jdbc:mysql://localhost:3306/department", "root",
        System.out.println("Connected");
        Statement st = connection.createStatement();
        ResultSet resultSet = st.executeQuery("SELECT * FROM `department`");
        while (resultSet.next()) {
            System.out.print("DNO = " + resultSet.getInt(1) + ", ");
            System.out.print("DNAME = " + resultSet.getString(2) + ", ");
            System.out.print("DADDRESS = " + resultSet.getString(3) + ", ");
            System.out.println("DPHONE = " + resultSet.getInt(4));
        }
        resultSet.close();
        st.close();
        connection.close();
    } catch (ClassNotFoundException | SQLException e) {
        e.printStackTrace();
    }
}
```

FIGURE – 9.5 Java code to connect MySQL database and to show the output.

The Output-



```
Connected
DNO = 10, DNAME = Marketing, DADDRESS = Dhaka, DPHONE = 783250936
DNO = 20, DNAME = IT, DADDRESS = Comilla, DPHONE = 783250946
DNO = 30, DNAME = HR, DADDRESS = Narayanganj, DPHONE = 783250938
DNO = 40, DNAME = Management, DADDRESS = Kulna, DPHONE = 783259936
DNO = 50, DNAME = Finance, DADDRESS = Sylhet, DPHONE = 757353226
BUILD SUCCESSFUL (total time: 3 seconds)
```

FIGURE – 9.6 Output of the query from the code.

DB Connection -3 :

Chandrima Dam (Table Name : **HOD**)

1. Choosing the IDE:

- For Java development, NetBeans IDE is selected.

2. Downloading MySQL Connector/J JDBC Driver:

- The MySQL Connector/J JDBC driver is downloaded from the official MySQL website, and the mysql-connector-j-8.4.0.jar file is extracted from the downloaded zip file.

3. Setting up XAMPP:

- XAMPP, a free and open-source web server solution stack developed by Apache Friends, is downloaded and installed.
- The XAMPP control panel is launched, ensuring that both the Apache and MySQL modules are running.
- The admin button beside the start column is clicked to open the PhpMyAdmin interface.

4. Creating a Database and Table in PhpMyAdmin:

- In the PhpMyAdmin interface, a new database named companymngt is created.
- Within the companymngt database, a new table named employee is created with columns HID (INT), Hname (VARCHAR), Hmail (VARCHAR), EPhn (INT), and Ejoindate (DATE). The SQL query to create the table is executed.

5. Setting up NetBeans IDE:

- NetBeans IDE is opened.
- Within the project, a package named employee is created.
- Inside the employee package, a Java class named Employee is created.
- The provided Java code is copied and pasted into the Employee.java file.

6. Adding MySQL Connector/J to the Project:

- The mysql-connector-j-8.4.0.jar file is added to the project's classpath:
- Right-click on the project name in the Project Explorer.
- Select "Properties" -> "Libraries" -> "Add JAR/Folder".
- Navigate to the location of the Connector/J JAR file and select it.

7. Updating MySQL Connection Information:

- If necessary, update the MySQL connection URL, username, and password in the DriverManager.getConnection() method call.

8. Running the Application:

- The HOD class is executed to run the application.

The table in mysql:

HID	HNAME	HMAIL	HPHN	HJOINDATE
15	SAGOR	cd@gmail.com	957534273	2002-02-12
25	FAREHA	fr@gmail.com	907346163	2014-03-30
35	SHUVRO	sk@gmail.com	980785098	2010-05-21
75	CHANDRIMA	cd@yahoo.com	980376383	2008-02-01

FIGURE – 9.7 Output of the table from the MySQL server.

The Main Code –

```
1 package hod;
2 import java.sql.Connection;
3 import java.sql.DriverManager;
4 import java.sql.ResultSet;
5 import java.sql.SQLException;
6 import java.sql.Statement;
7 public class Hod {
8     public static void main(String[] args) {
9         try {
10             Class.forName("com.mysql.cj.jdbc.Driver");
11             Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/companymngt", "root", "");
12             System.out.println("Connected");
13             Statement st = conn.createStatement();
14             ResultSet resultSet = st.executeQuery("SELECT * FROM hod;");
15             while (resultSet.next()) {
16                 System.out.print ("HID = " + resultSet.getInt(1) + ", ");
17                 System.out.print ("HName = " + resultSet.getString(2) + ", ");
18                 System.out.print ("Hmail = " + resultSet.getString(3) + ", ");
19                 System.out.println("Hphn = " + resultSet.getInt(4) + ", ");
20                 System.out.println("Hjoindate = " + resultSet.getDate(5));
21             }
22             conn.close();
23         } catch (Exception s) {
24             System.out.println(s);
25         }
26     }
27 }
```

FIGURE – 9.8 Java code to connect MySQL database and to show the output.

The Output-

```
HID    = 15, HName = SAGOR, Hmail = cd@gmail.com, Hphn  = 957534273,
Hjoindate = 2002-02-12
HID    = 25, HName = FAREHA, Hmail = fr@gmail.com, Hphn  = 907346163,
Hjoindate = 2014-03-30
HID    = 35, HName = SHUVRO, Hmail = sk@gmail.com, Hphn  = 980785098,
Hjoindate = 2010-05-21
HID    = 75, HName = CHANDRIMA, Hmail = cd@yahoo.com, Hphn  = 980376383,
Hjoindate = 2008-02-01
BUILD SUCCESSFUL (total time: 0 seconds)
```

FIGURE – 9.9 Output of the query from the code.

DB Connection -4:

Shuvra Kanti Das (Table Name : **Employee**)

1. An IDE such as Eclipse, IntelliJ IDEA, or NetBeans is chosen for Java development. For this case study, NetBeans IDE is used.
2. The MySQL Connector/J JDBC driver is downloaded from the official MySQL website, and the downloaded zip file is extracted to find the **mysql-connector-j-8.4.0.jar** file.
3. XAMPP, a free and open-source cross-platform web server solution stack package developed by Apache Friends, is downloaded, and installed.
4. The XAMPP control panel is started, and it is ensured that the Apache and MySQL modules are running. The admin button right beside the start column is pressed.
5. A web browser is opened, and **http://localhost/phpmyadmin** is navigated to access the PhpMyAdmin interface. In the PhpMyAdmin interface, a new database named **companymngt** is created. Within the **companymngt** database, a new table named **employee** is created with the following columns: **EID** (INT), **Ejoindate** (DATE), **Ename** (VARCHAR), **Email** (VARCHAR), **Esal** (INT), **Eadress** (VARCHAR). Then, the SQL query to create the table is executed.

The table in MySQL:

EID	EJOINDATE	ENAME	EMAIL	ESAL	EADDRESS
30	2008-03-01	ALEX CARRY	x@gmail.com	12000	BARISAL
40	2018-09-23	MARK WOOD	a@gmail.com	30000	MOULVIBAZAR
50	2009-08-10	TIM DAVID	t@gmail.com	4000	CHITTAGONG
60	2015-11-17	MARRY COM	m@gmail.com	75000	MYMENSINGH
70	2023-04-30	PAT MILLS	p@gmail.com	6000	GAZIPUR

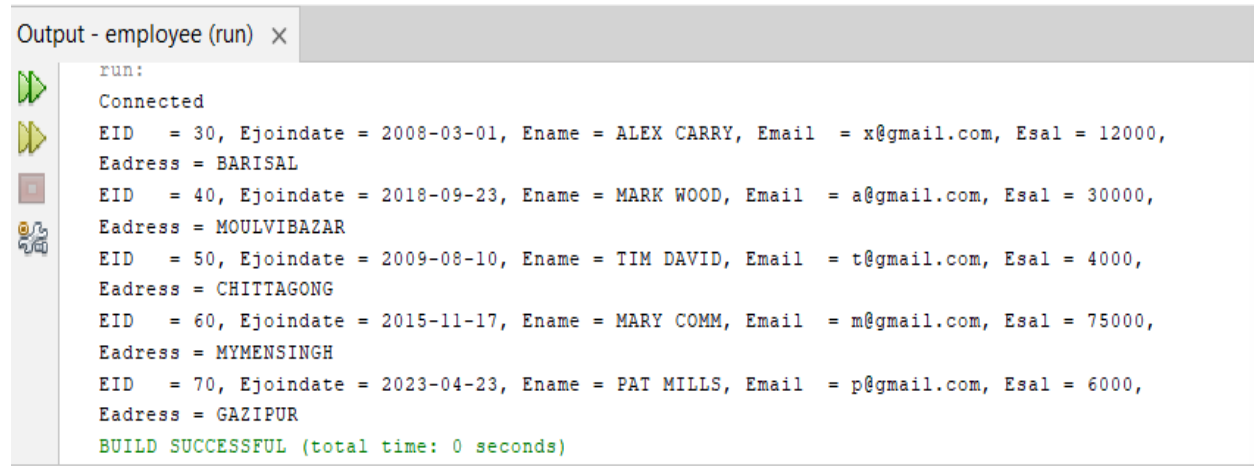
FIGURE – 9.10 Output of the table from the MySQL server.

The Main Code –

```
1  package employee;
2  import java.sql.Connection;
3  import java.sql.DriverManager;
4  import java.sql.ResultSet;
5  import java.sql.SQLException;
6  import java.sql.Statement;
7
8  public class Employee {
9      public static void main(String[] args) {
10         try {
11             Class.forName("com.mysql.cj.jdbc.Driver");
12             Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/companymngt", "root", "");
13             System.out.println("Connected");
14             Statement st = conn.createStatement();
15             ResultSet resultSet = st.executeQuery("SELECT * FROM employee;");
16             while (resultSet.next()) {
17                 System.out.print("EID = " + resultSet.getInt(1) + ", ");
18                 System.out.print("Ejoindate = " + resultSet.getDate(2) + ", ");
19                 System.out.print("Ename = " + resultSet.getString(3) + ", ");
20                 System.out.print("Email = " + resultSet.getString(4) + ", ");
21                 System.out.println("Esal = " + resultSet.getInt(5) + ", ");
22                 System.out.println("Eadress = " + resultSet.getString(6));
23             }
24             conn.close();
25         } catch (Exception e) {
26             System.out.println(e);
27         }
28     }
29 }
30
```

FIGURE – 9.11 Java code to connect MySQL database and to show the output.

The Output-



```
Output - employee (run) x
run:
Connected
EID   = 30, Ejoindate = 2008-03-01, Ename = ALEX CARRY, Email  = x@gmail.com, Esal = 12000,
Eadress = BARISAL
EID   = 40, Ejoindate = 2018-09-23, Ename = MARK WOOD, Email  = a@gmail.com, Esal = 30000,
Eadress = MOULVIBAZAR
EID   = 50, Ejoindate = 2009-08-10, Ename = TIM DAVID, Email  = t@gmail.com, Esal = 4000,
Eadress = CHITTAGONG
EID   = 60, Ejoindate = 2015-11-17, Ename = MARY COMM, Email  = m@gmail.com, Esal = 75000,
Eadress = MYMENSINGH
EID   = 70, Ejoindate = 2023-04-23, Ename = PAT MILLS, Email  = p@gmail.com, Esal = 6000,
Eadress = GAZIPUR
BUILD SUCCESSFUL (total time: 0 seconds)
```

FIGURE – 9.12 Output of the query from the code.

5.The NetBeans IDE is opened, and within the project, a package named **employee** is created. Inside the **employee** package, a Java class named **Employee** is created. The provided Java code is copied and pasted into the **Employee.java** file. The MySQL Connector/J JAR file (**mysql-connector-j-8.4.0.jar**) is added to the project's class path:

6.The project name in the Project Explorer is right clicked. "Properties" -> "Libraries" -> "Add JAR/Folder" is selected. The location where the Connector/J JAR file is found is navigated to, and it is selected.

7.The MySQL connection URL, username, and password in the **DriverManager.getConnection()** method call are updated if necessary.

8.The **Employee** class is run.

Conclusion

In this project, we have done normalization and finalization from an ER diagram named “Company Structure Management System” which had five different relations among the entities. After getting the final tables we showed all the queries that are necessary to create those tables in our own schema. We also showed how to insert values in those tables and took their screenshots. After that we did some query tests. At last, we have created some tables in mysql server and connected it with an IDE to show that these tables can be accessed by a java code.